

Grade 8
Unit 12
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. Find the value of the expression $\frac{7.2 + \sqrt[3]{w}}{20}$, where $w = -125$.

$$\frac{7.2 + \sqrt[3]{-125}}{20} = \frac{7.2 - 5}{20} = \frac{2.2}{20} = \boxed{0.11}$$

2. Find the value of the expression $\frac{\sqrt{81-x^2}}{4} + 19$, where $x = 5$.

$$\frac{\sqrt{81} - 5^2}{4} + 19 \rightarrow \frac{9 - 25}{4} + 19 \rightarrow \frac{-16}{4} + 19 \rightarrow -4 + 19 = \boxed{15}$$

3. Find the value of the expression $\sqrt{a^2 - 144}$, where $a = -13$.

$$\sqrt{(-13)^2 - 144} \rightarrow \sqrt{169 - 144} \rightarrow \sqrt{25} = \boxed{5}$$

4. Find the value of the expression $\frac{7 \cdot y^3}{-\sqrt{64}}$, where $y = -2$.

$$\frac{7 \cdot (-2)^3}{-\sqrt{64}} \rightarrow \frac{7(-8)}{-8} \rightarrow \frac{-56}{-8} = \boxed{7}$$

For questions 5-10, find the value of each expression.

$$5. \quad \left| 10^{-1} - \sqrt[3]{\frac{1}{125}} \right| \rightarrow \left| \frac{1}{10} - \frac{1}{5} \right| \rightarrow \left| \frac{1}{10} - \frac{2}{10} \right| \rightarrow \left| -\frac{1}{10} \right| = \boxed{\frac{1}{10}}$$

$$6. \quad \frac{30 - 4 \cdot \sqrt{121}}{3^2} \rightarrow \frac{30 - 4(11)}{9} \rightarrow \frac{30 - 44}{9} = \boxed{-\frac{14}{9} \text{ or } -1\frac{5}{9}}$$

$$7. \quad \frac{2 \cdot 3^3}{-\sqrt{36}} \quad \frac{2 \cdot 27}{-6} \rightarrow \frac{54}{-6} = \boxed{-9}$$

$$8. \quad \left(\frac{1}{4}\right)^{-2} + \sqrt{5^2 - 4^2} \rightarrow \frac{4^2}{1^2} + \sqrt{25 - 16} \rightarrow \frac{16}{1} + \sqrt{9} \rightarrow 16 + 3 = \boxed{19}$$

$$9. \quad \frac{10.5 + \sqrt{225}}{30} \rightarrow \frac{10.5 + 15}{30} \rightarrow \frac{25.5}{30} = \boxed{0.85}$$

$$10. \quad \left| \left(\frac{1}{2}\right)^{-2} + \sqrt[3]{\frac{1}{27}} \right|$$

$$\rightarrow \left| \frac{2^2}{1^2} + \frac{1}{3} \right| \rightarrow \left| \frac{4}{1} + \frac{1}{3} \right| \rightarrow \left| \frac{12}{3} + \frac{1}{3} \right| \rightarrow \left| \frac{13}{3} \right| = \boxed{\frac{13}{3} \text{ or } 4\frac{1}{3}}$$